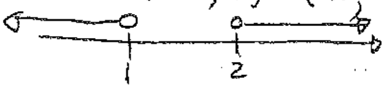
WTW 158 Study guide solutions. (Stewart)

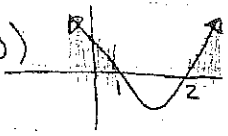
Unit 1.1

Ex. A p A9

25. $(x-1)(x-2) > 0$

$\therefore x \in (-\infty, 1) \cup (2, \infty)$



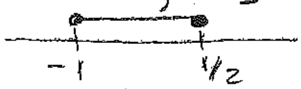


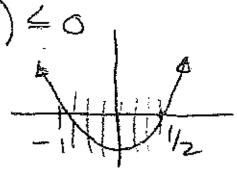
27. $2x^2 + x \leq 1$

$\Rightarrow 2x^2 + x - 1 \leq 0$

$\Rightarrow (2x-1)(x+1) \leq 0$

$x \in [-1, \tfrac{1}{2}]$





42. $h \geq 15$

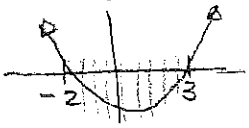
$\Rightarrow 30 + 10t - 5t^2 \geq 15$

$\Rightarrow 5t^2 - 10t - 15 \leq 0$

$\Rightarrow 5(t-3)(t+2) \leq 0$

$t \in [-2, 3]$

but $t \geq 0$



$\Rightarrow$ ball at least 15 m above the ground in the time interval $[0, 3]$.

58. $a \leq bx + c < 2a$

$\Rightarrow a - c \leq bx < 2a - c$

$\Rightarrow \frac{a-c}{b} \leq x < \frac{2a-c}{b}$

since $b > 0$

59. $ax + b < c$

$\Rightarrow ax < c - b$

$\Rightarrow x > \frac{c-b}{a}$

since $a < 0$.

60. $\frac{ax+b}{c} \leq b$

$\Rightarrow ax + b \geq bc$ (since $c < 0$)

$\Rightarrow ax \geq bc - b$

$\Rightarrow x \leq \frac{b(c-1)}{a}$ (since $a < 0$)

Unit 1.2

Ex A p A9

1. $|5 - 23| = |-18| = 18$

3. $|-\pi| = \pi$ $(\pi > 0)$

7. $|x-2|$ if $x < 2$

$$= \begin{cases} x-2 & \text{if } x-2 \geq 0 \quad (x \geq 2) \\ -(x-2) & \text{if } x-2 < 0 \quad (x < 2) \end{cases}$$

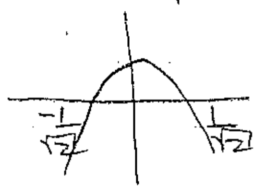
So $|x-2| = -x + 2$ if $x < 2$

12. $$|1-2x^2| = \begin{cases} 1-2x^2 & \text{if } 1-2x^2 \geq 0 \\ -1+2x^2 & \text{if } 1-2x^2 < 0 \end{cases}$$

Consider $1 - 2x^2 = 0$

$\Rightarrow x = \pm \frac{1}{\sqrt{2}}$

Parabola with x-intercepts $\pm \frac{1}{\sqrt{2}}$



So $1 - 2x^2 \geq 0 \Rightarrow -\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}}$

and $1 - 2x^2 < 0 \Rightarrow x < -\frac{1}{\sqrt{2}}$ or $x > \frac{1}{\sqrt{2}}$

Therefor:

$$|1-2x^2| = \begin{cases} 1-2x^2 & \text{if } -\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}} \\ -1+2x^2 & \text{if } x < -\frac{1}{\sqrt{2}} \text{ or } x > \frac{1}{\sqrt{2}} \end{cases}$$

56. $0 < |x-5| < \frac{1}{2}$

$\Rightarrow |x-5| > 0$ and $|x-5| < \frac{1}{2}$

This is true for all $x \neq 5$

$-\frac{1}{2} < x-5 < \frac{1}{2}$

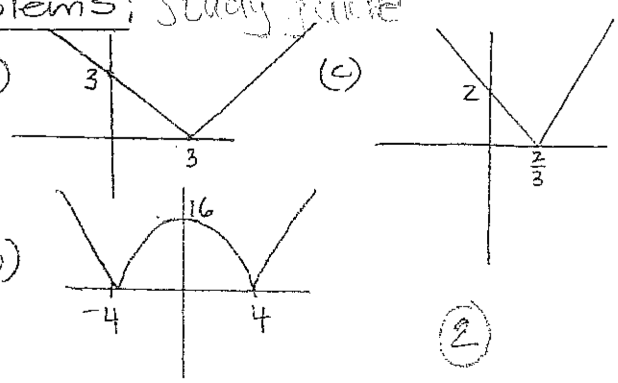
$4.5 < x < 5.5$

So $x \in (-4.5, 5) \cup (5, 5.5)$

Problems; Study guide

2 (a)   (c)

(b)



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Unit 1.3

Ex 1.1 p19

26. A balloon with radius $r+1$ has volume $V(r+1) = \frac{4}{3}\pi(r+1)^3$
$= \frac{4}{3}\pi(r^3+3r^2+3r+1)$

$\Rightarrow V(r+1) - V(r)$
$= 4/3\,\pi\,(3r^2+3r+1)$

34. $g(t) = \sqrt{3-t} - \sqrt{2+t}$ is defined when

$3-t \geq 0$ and $2+t \geq 0$

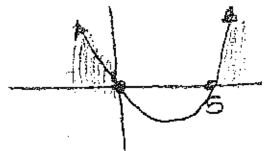
$\Rightarrow t \leq 3$ and $t \geq -2$

$t \in [-2, 3]$

35. $h(x) = \dfrac{1}{\sqrt[4]{x^2-5x}}$ is defined

if $x^2-5x > 0$

$\Rightarrow x(x-5) > 0$



$\Rightarrow x \in (-\infty, 0) \cup (5, \infty)$

38. $h(x) = \sqrt{4-x^2}$

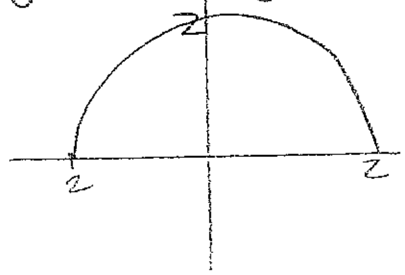
Say $y = \sqrt{4-x^2}$

$\Rightarrow y^2 = 4-x^2 \Rightarrow x^2+y^2 = 4$

so $h(x)$ is half a circle with radius 2.

Domain is $\{x \mid -2 \leq x \leq 2\}$

Range is $\{y \mid 0 \leq y \leq 2\}$



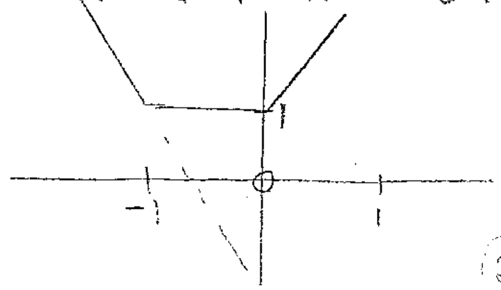
48. $h(t) = |t| + |t+1|$ and

$|t| = \begin{cases} t & \text{if } t \geq 0 \\ -t & \text{if } t < 0 \end{cases}$ and

$|t+1| = \begin{cases} t+1 & \text{if } t \geq -1 \\ -t-1 & \text{if } t < -1 \end{cases}$ so

$h(t) = \begin{cases} t+(t+1) & \text{if } t \geq 0 \\ -t+(t+1) & \text{if } -1 \leq t < 0 \\ -t+(-t-1) & \text{if } t < -1 \end{cases}$

$= \begin{cases} 2t+1 & \text{if } t \geq 0 \\ 1 & \text{if } -1 \leq t < 0 \\ -2t-1 & \text{if } t < -1 \end{cases}$



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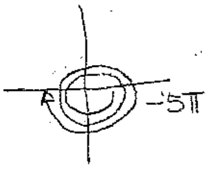
Unit 1·4 Ex D p A32

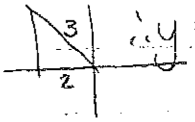
26. 

$x=-1,\ y=0,\ r=1$
$\cos(-5\pi) = -1$
$\tan(-5\pi)$ undef etc.

31. $\sec\phi = -\frac{3}{2}$ $\therefore$ $r=3,\ x=-2$

$\therefore \sin\phi = \frac{\sqrt{5}}{3}$

$\tan\phi = -\frac{\sqrt{5}}{2}$

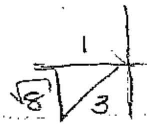


$\therefore y = \sqrt{5}$

32. $\cos x = -\frac{1}{3}$ $\therefore$ $r=3,\ x=-1$

$\sin x = -\frac{\sqrt{8}}{3}$

$\tan x = \sqrt{8}$

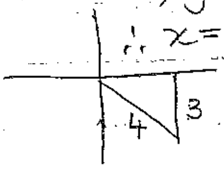


$\therefore y = -\sqrt{8}$

34. $\operatorname{cosec}\theta = -\frac{4}{3}$ $\therefore$ $r=4,\ y=-3$

$\cos\theta = \frac{\sqrt{7}}{4}$

$\tan\theta = -\frac{3}{\sqrt{7}}$



$\therefore x = \sqrt{7}$

37. $\tan\frac{2\pi}{5} = \frac{x}{8} \Rightarrow x = 8\tan\frac{2\pi}{5}$

$= 24.62\text{cm}$

72. $2+\cos 2x = 3\cos x$

$2+2\cos^2 x - 1 - 3\cos x = 0$

$2\cos^2 x - 3\cos x + 1 = 0$

$(2\cos x - 1)(\cos x - 1) = 0$

$\Rightarrow \cos x = \frac{1}{2}$ or $\cos x = 1$

$\Rightarrow x = \frac{\pi}{3}, \frac{5\pi}{3}$ or $x = 0, 2\pi$

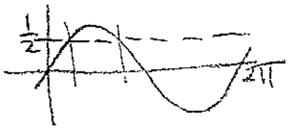
So $x = \frac{\pi}{3}, \frac{5\pi}{3}, 0, 2\pi.$

73. $\sin x \le \frac{1}{2}$

$\sin x = \frac{1}{2}$

when $x = \frac{\pi}{6}$ or

$x = \frac{5\pi}{6}$



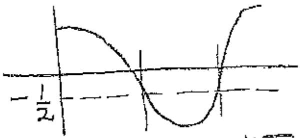
$\therefore \sin x \le \frac{1}{2} \Rightarrow$

$0 \le x \le \frac{\pi}{6}$ or $\frac{5\pi}{6} \le x \le 2\pi$

for $x \in [0, 2\pi]$

74. $2\cos x + 1 > 0$

$\cos x > -\frac{1}{2}$



$\cos x = -\frac{1}{2}$ when $x = \frac{2\pi}{3}$ or $\frac{4\pi}{3}$

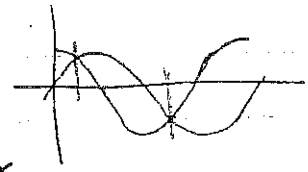
$\Rightarrow \cos x > -\frac{1}{2}$ when $0 < x < \frac{2\pi}{3}$

or $\frac{4\pi}{3} < x \le 2\pi$

76. $\sin x > \cos x$

$\sin x = \cos x \Rightarrow \tan x = 1$

when $x = \frac{\pi}{4}$ or $x = \frac{5\pi}{4}$



$\sin x > \cos x$

when $\frac{\pi}{4} < x < \frac{5\pi}{4}$

2. $\sin(75\pi) = \sin(\pi) = 0$

$\cos\left(\frac{17\pi}{4}\right) = \cos\left(\frac{16\pi}{4} + \frac{\pi}{4}\right)$

$= \cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}$

Unit 1.5

Ex 1.2 p33.

1. (a) Logarithmic function
   (b) Root function, $n=4$
   (c) Rational function
   (d) Polynomial degree 2.
   (e) Exponential function
   (f) Trigonometric function.

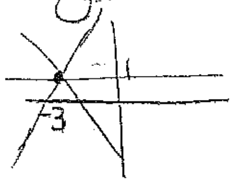
2. (a) Exponential function.
   (b) Power function.
   (c) Polynomial degree 5
   (d) Trigonometric
   (e) Rational function.
   (f) Algebraic function

3. (a) h (b) f (c) g

4. (a) G (b) f (c) F (d) g

6. $1-\tan x \neq 0 \Rightarrow \tan x \neq 1$
   $\Rightarrow x \neq \frac{\pi}{4} + n\pi$ and
   $\tan x$ is not defined if
   $x = \frac{\pi}{2} + n\pi$
   $\Rightarrow$ Domain is $\{x \mid x \neq \frac{\pi}{4} + n\pi, x \neq \frac{\pi}{2} + n\pi, n \in \mathbb{Z}\}$

8. $f(x) - 1 = m(x+3)$ are lines passing through the point $(-3, 1)$



11. $f(-1) = f(0) = f(2) = 0$
   $f(x) = a(x+1)x(x-2)$
   $= ax(x+1)(x-2)$ passing through $(1, 6)$
   $\Rightarrow 6 = a(1)(2)(-1)$
   $\Rightarrow -2a = 6 \Rightarrow a = -3$
   $\Rightarrow f(x) = -3x(x+1)(x-2)$

12. $T = 0.02t + 8.50$
   (a) The slope is 0.02 – average surface temperature is increasing by 0.02 °C per year
   The T-intercept is 8.50 – represents the average surface temperature at $t=0$ that is in 1900.
   (b) In 2100 $t = 200$
   $\Rightarrow T(200) = 0.02(200) + 8.50$
   $= 12.50\,°C$

13. $D = 200$ mg so
   $c = 0.0417(200)(a+1)$
   $= 8.34a + 8.34$
   (a) Slope is 8.34 – the change in mg of dosage for a child for each change of 1 year in age.
   (b) For a newborn $a=1$
   so dosage $c = 8.34$ mg.

Unit 1.6

Ex 1.3 p42

1. (a) $y = f(x) + 3$
   (b) $y = f(x) - 3$
   (c) $y = f(x-3)$
   (d) $y = f(x+3)$
   (e) $y = -f(x)$
   (f) $y = f(-x)$
   (g) $y = 3f(x)$
   (h) $y = \frac{1}{3}f(x)$

3. (a) graph 3 (b) graph 1
   (c) graph 4 (d) graph 5
   (e) graph 2

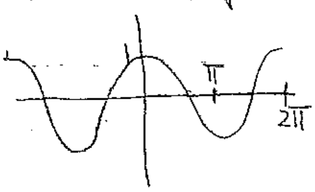
6. Shifted 2 units to right with vertical stretch of factor 2.

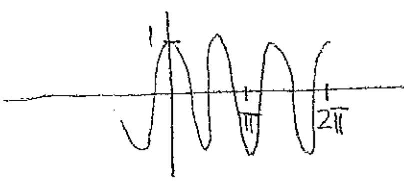
$$y = 2f(x-2) = 2\sqrt{3(x-2)-(x-2)^2}$$
$$= 2\sqrt{-x^2+7x-10}$$

7. Shifted 4 units left, reflected about x-axis, 1 unit down.

$$y = -f(x+4) - 1$$
$$= -\sqrt{-x^2-5x-4} - 1$$

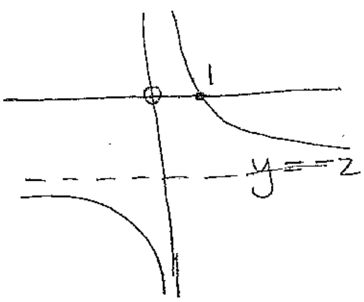
13.





Compress horizontally by factor 3.

20.



32. (a) $(f+g)(x)$
$= \sqrt{3-x} + \sqrt{x^2-1}$

$D = (-\infty,-1] \cup [1,3]$

(b) $(f-g)(x)$
$= \sqrt{3-x} - \sqrt{x^2-1}$

$D = (-\infty,-1] \cup [1,3]$

(c) $(fg)(x) = \sqrt{3-x} \cdot \sqrt{x^2-1}$

$D = (-\infty,-1] \cup [1,3]$

(d) $\left(\frac{f}{g}\right)(x) = \sqrt{3-x}/\sqrt{x^2-1}$

$D = (-\infty,-1) \cup (1,3]$

37. $f(x) = x + \frac{1}{x}$

$D_f = \{x \mid x \neq 0\}$

$g(x) = \frac{x+1}{x+2}$

$D_g = \{x \mid x \neq -2\}$

(a) $f(g(x)) = f\left(\frac{x+1}{x+2}\right)$ $x \neq -2$

$= \frac{x+1}{x+2} + \frac{x+2}{x+1}$ $x \neq -1$

$D_{f\circ g} = \{x \mid x \neq -1, -2\}$

(b) $g(f(x)) = g\left(x + \frac{1}{x}\right)$ $x \neq 0$

$= \dfrac{x + \frac{1}{x} + 1}{x + \frac{1}{x} + 2}$

$= \dfrac{x^2+x+1}{(x+1)^2}$ $x \neq -1$

$D_{g\circ f} = \{x \mid x \neq -1, 0\}$

(c) $f(f(x)) = f\left(x + \frac{1}{x}\right)$ $x \neq 0$

$= x + \frac{1}{x} + \dfrac{1}{x + 1/x}$

$= \dfrac{x^4+3x^2+1}{x(x^2+1)}$

$D_{f\circ f} = \{x \mid x \neq 0\}$

(d) $g(g(x)) = g\left(\frac{x+1}{x+2}\right) \quad x \neq -2$

$$= \frac{\frac{x+1}{x+2}+1}{\frac{x+1}{x+2}+2} = \frac{2x+3}{3x+5} \qquad x \neq -\frac{5}{3}$$

$D_{g\circ g} = \{x \mid x \neq -2, -\frac{5}{3}\}$

45. $F(x) = \dfrac{\sqrt[3]{x}}{1+\sqrt[3]{x}}$

$g(x) = \sqrt[3]{x}, \quad f(x) = \dfrac{x}{1+x}$

$f(g(x)) = f(\sqrt[3]{x}) = \dfrac{\sqrt[3]{x}}{1+\sqrt[3]{x}}$

46. $G(x) = \sqrt[3]{\dfrac{x}{1+x}}$

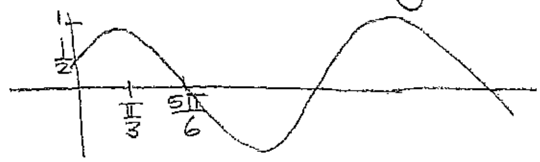
$g(x) = \dfrac{x}{1+x}, \quad f(x) = \sqrt[3]{x}$

$f(g(x)) = f\left(\dfrac{x}{1+x}\right)$

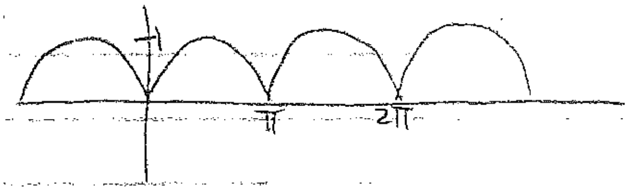
$= \sqrt[3]{\dfrac{x}{1+x}} = G(x)$

Ex D pA32

77. Start with $y = \cos x$ and shift it $\frac{\pi}{3}$ to the right.

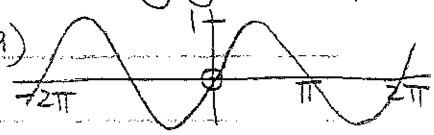


81. Start with $y = \sin x$ and reflect the parts below the x-axis about the x-axis.



3. Study guide problems.

(a)



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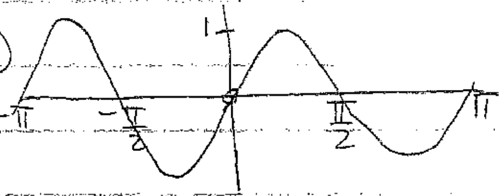
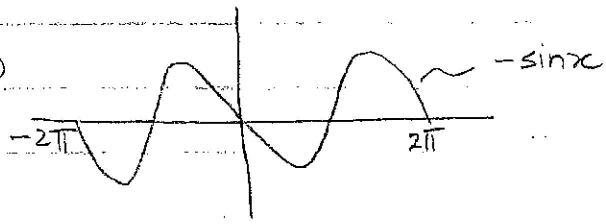
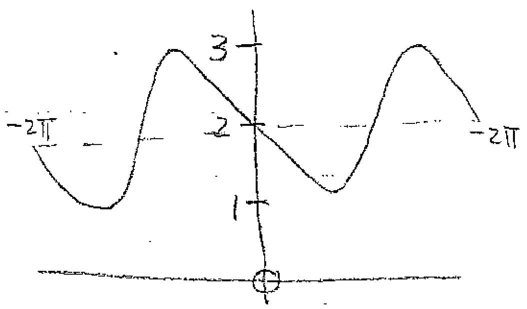


(c)



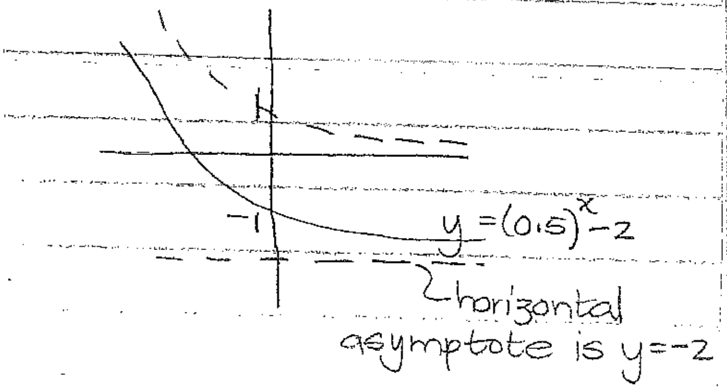


Unit 1.7

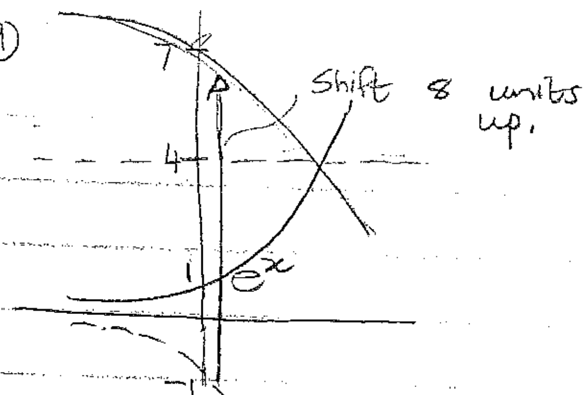
Ex 1.4 p53

1. (a) $\frac{4^{-3}}{2^{-8}} = \frac{2^8}{4^3} = \frac{2^8}{2^6} = 2^2 = 4$

14. $y = (0.5)^x - 2$
Start with $y = \left(\frac{1}{2}\right)^x = 2^{-x}$
and shift two unit down.

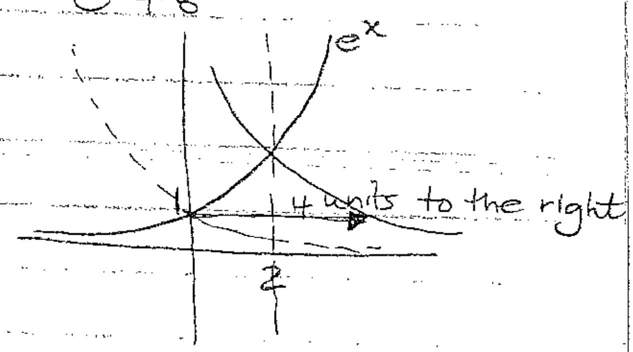


18. (a)



Reflect graph about the x-axis so $y = -e^x$, then shift graph up by 8 units.
$y = -e^x + 8$

(b)



Reflect e^x about the y-axis so $y = e^{-x}$ then shift 4 units to the right
$y = e^{-(x-4)}$

21. $y = Cb^x$ through points: $(1, 6)$ and $(3, 24)$

$\therefore\ 6 = Cb^1$ and $24 = Cb^3$

$\Rightarrow\ C = \frac{6}{b}$ $\Rightarrow$ $24 = \left(\frac{6}{b}\right)b^3$

$24 = 6b^2$

$b^2 = 4$

$b = 2$

since $b > 0$ and $\Rightarrow C = 3$

So $y = 3(2)^x$

30 (a) One doubling period is 30 min, so 6 doubling periods in 3 hours.

(b) In t hours there will be $2t$ doubling periods.
Population y at time t is $y = 500 \cdot 2^{2t}$

(c) $t = 40\text{min} = \frac{40}{60} = \frac{2}{3}\text{hrs}$
$y = 500 \cdot 2^{2\left(\frac{2}{3}\right)}$
≈ 1260

Unit 1.8

Ex 1.5 p66.

5. Function not one-to-one. Fails horizontal line test.

7. Function is one-to-one. No horizontal line will intersect twice.

15. (a) f 1-1 so $f(4) = 7$
$\Rightarrow\ 4 = f^{-1}(7)$

(b) f 1-1 so $f^{-1}(8) = 2$
$\Rightarrow\ 8 = f(2)$

18. (a) Pass horizontal line test

(b) $D_f = [-3, 3] = R_{f^{-1}}$

$D_f = [-1, 3] = D_{f^{-1}}$

(c) $f(0) = 2 \Rightarrow f^{-1}(2) = 0$

(d) $f(-1.7) \approx 0 \Rightarrow f^{-1}(0) \approx -1.7$

21. $f(x) = 1 + \sqrt{2+3x}$

Let $y = 1 + \sqrt{2+3x}$ consider

$x = 1 + \sqrt{2+3y}$

$\Rightarrow (x-1)^2 = 2 + 3y$

$\Rightarrow y = \frac{1}{3}(x-1)^2 - \frac{2}{3}$

Note: $R_f = [1, \infty)$

so $D_{f^{-1}} = [1, \infty)$

Unit 1.9

Ex 1.5 p66

23. $f(x) = e^{2x-1}$

Let $y = e^{2x-1}$ so consider

$x = e^{2y-1}$

$\Rightarrow \ln x = 2y - 1$

$\Rightarrow y = \frac{1}{2}(\ln x + 1)$

25. $y = \ln(x+3)$

Consider $x = \ln(y+3)$

$\Rightarrow e^x = y + 3$

$\Rightarrow y = e^x - 3$

36. (a) $\log_5 \frac{1}{125} = \log_5 5^{-3}$

$= -3$

(b) $\ln\left(\frac{1}{e^2}\right) = \ln 1 - \ln e^2$

$= -2$

38. (a) $e^{-\ln 2} = e^{\ln 2^{-1}} = 2^{-1} = \frac{1}{2}$

(b) $e^{\ln(\ln e^3)}$

$= e^{\ln(3 \ln e)} = e^{\ln 3} = 3$

52. (a) $\ln(x^2-1) = 3$

$\Rightarrow x^2 - 1 = e^3$

$\Rightarrow x^2 = e^3 + 1$

$\therefore x = \pm\sqrt{e^3+1}$

(b) $e^{2x} - 3e^x + 2 = 0$

$(e^x - 1)(e^x - 2) = 0$

$\Rightarrow e^x = 1$ or $e^x = 2$

$\Rightarrow x = \ln 1$ or $x = \ln 2$

$\therefore x = 0$

54. (a) $\ln(\ln x) = 1$

$\ln x = e^1$

$x = e^e$

(b) $e^{ax} = Ce^{bx}$, $a \neq b$

$\ln e^{ax} = \ln(Ce^{bx})$

$ax = \ln C + \ln e^{bx}$

$ax = \ln C + bx$

$\Rightarrow x(a-b) = \ln C$

$\Rightarrow x = \frac{\ln C}{a-b}$

56. (a) $1 < e^{3x-1} < 2$

$\Rightarrow \ln 1 < 3x - 1 < \ln 2$

$\Rightarrow 1 < 3x < \ln 2 + 1$

$\Rightarrow \frac{1}{3} < x < \frac{1}{3}(\ln 2 + 1)$

(b) $1 - 2\ln x < 3$

$-2\ln x < 2$

$\ln x > -1$

$\Rightarrow x > e^{-1}$ or

$x > \frac{1}{e}$

Unit 1·10

Ex 1·5 p66

63. (a) $\cos^{-1}(-1) = \pi$ since $\cos \pi = -1$

(b) $\sin^{-1}(0.5) = \frac{\pi}{6}$ since $\sin \frac{\pi}{6} = \frac{1}{2}$

64. (a) $\tan^{-1}\sqrt{3} = \frac{\pi}{3}$ since $\tan\frac{\pi}{3} = \sqrt{3}$

(b) $\arctan(-1) = -\frac{\pi}{4}$ since $\tan\left(-\frac{\pi}{4}\right) = -1$

65. (a) $\operatorname{cosec}^{-1}(\sqrt{2}) = \frac{\pi}{4}$ since $\operatorname{cosec}\frac{\pi}{4} = \sqrt{2}$

(b) $\arcsin 1 = \frac{\pi}{2}$ since $\sin\frac{\pi}{2} = 1$

66. (a) $\sin^{-1}\left(-\frac{1}{\sqrt{2}}\right) = -\frac{\pi}{4}$

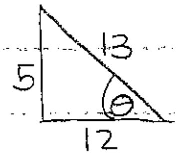
(b) $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}$

68. (a) $\arcsin\left(\sin\left(\frac{5\pi}{4}\right)\right)$
$= \arcsin\left(-\frac{1}{\sqrt{2}}\right) = -\frac{\pi}{4}$

(b) $\cos\left(2\sin^{-1}\left(\frac{5}{13}\right)\right)$

Let $\theta = \sin^{-1}\left(\frac{5}{13}\right)$

$\Rightarrow \sin\theta = \frac{5}{13}$



So that

$\cos\left(2\sin^{-1}\left(\frac{5}{13}\right)\right)$

$= \cos 2\theta = \cos^2\theta - \sin^2\theta$

$= \left(\frac{12}{13}\right)^2 - \left(\frac{5}{13}\right)^2$

$= \frac{119}{169}$

69. Let $y = \sin^{-1}x$

$\Rightarrow \sin y = x$ and $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

and $\cos^2 y + \sin^2 y = 1$

$\Rightarrow \cos y = +\sqrt{1-\sin^2 y}$ as y is in 1st or 4th quadrant

$\Rightarrow \cos y = \sqrt{1-x^2}$

$\therefore \cos(\sin^{-1}x) = \sqrt{1-x^2}$

70. $\tan(\sin^{-1}x)$

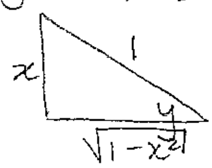
Let $y = \sin^{-1}x$

$\Rightarrow \sin y = x$ and $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

So $\tan(\sin^{-1}x)$

$= \tan y$

$= x/\sqrt{1-x^2}$



71. $\sin(\tan^{-1}x)$

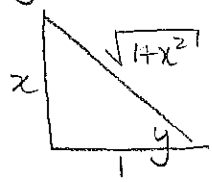
Let $y = \tan^{-1}x$

$\Rightarrow \tan y = x$ and $y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

So $\sin(\tan^{-1}x)$

$= \sin y$

$= x/\sqrt{1+x^2}$

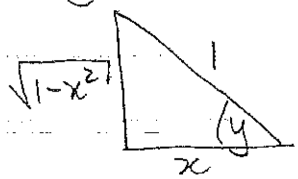


72. $\sin(2\arccos x)$

Let $y = \arccos x$

$\Rightarrow \cos y = x$ and $y \in [0, \pi]$

So



$\sin(2\arccos x)$

$= \sin(2y)$

$= 2\sin y\cos y$

$= 2\left(\sqrt{1-x^2}\right)(x) = 2x\sqrt{1-x^2}$

75. $g(x) = \sin^{-1}(3x+1)$

$\Rightarrow \sin g(x) = 3x+1$ and

$-\frac{\pi}{2} \le g(x) \le \frac{\pi}{2}$ and $-1 \le 3x+1 \le 1$

$\Rightarrow -2 \le 3x \le 0$

$\Rightarrow -\frac{2}{3} \le x \le 0$

$D_g = \left[-\frac{2}{3}, 0\right]$

$R_g = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$